TS2000
TS2100
Robot Controller

FIELD BUS SLAVE FUNCTION MANUAL

Rev.8
April 1, 2020

Notice

1. Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
2. Before operating the industrial robot, read through and completely understand this manual.
3. After reading through this manual, keep it nearby for future reference.

Shibaura Machine

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The information contained in this manual is subject to change without notice to effect improvements.

Preface

This manual describes the initial setting procedures and operating method of the field bus slave function, the TS series robot controller extension function. It is intended for users with general knowledge of field bus.
The slave module for the field bus can select either of the three (3) types; Profibus, DeviceNet or CC–Link, which should be determined at order entry.
The field bus slave function uses the AnyBus–S board made by HMS.
For details on the AnyBus–S board, refer to the following web site.

- Profibus
  Uses AnyBus–S Profibus–DP made by HMS.
- DeviceNet
  Uses AnyBus–S DeviceNet Parallel made by HMS.
- CC–Link
  Uses AnyBus–S CC–LINK made by HMS.

*1: HMS's internet web site

Homepage written in Japanese
http://www.hms-networks.co.jp/

Homepage written in English
http://www.hms-networks.com/index.html

Japanese download site
http://www.anybus.com/jpweb/downloads/absdownloads.shtml

English download site
http://www.hms-networks.com/downloads/absdownloads.shtml

*2: Instruction manuals which are referred to from this manual

- Robot language manual
- Operation manual
- User parameter manual
- Maintenance manual
- Instruction manual of master station you wish to use

! **CAUTION**	This manual does not contain any detailed descriptions on power and robot connection. For the connection of the power and robot, see the Installation & Transport Manual.

Cautions on Safety

This manual contains the important information on the robot and controller to prevent injury to the operators and persons nearby, to prevent damages to assets and to assure correct use.
Make sure that you well understand the following details (indications and symbols) before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
! **DANGER**	This means that "incorrect handling will lead to fatalities or serious injuries."
! **CAUTION**	This means that "incorrect handling may lead to personal injuries 1) or physical damage 2)."

1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long-term medical treatment.

2) Physical damage refers to damages due to destruction of assets or resources.

[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited (must not be done). Details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory (must be done). Details of the actions that must be done are indicated with pictures or words in or near the symbol.
	This means danger. Details of the actual danger are indicated with pictures or words in or near the symbol.
	This means caution. Details of the actual caution are indicated with pictures or words in or near the symbol.

! **CAUTION**	To perform the work ranging from robot installation to operation with safety, read through and through the Safety Manual provided separately before actually starting the work.

Maintenance and Inspection

To use the robot safety, strictly observe the following matters.

! DANGER	
 Prohibited	• NEVER burn, disassemble or charge the battery. Otherwise, it may explode.
 Mandatory	• Before performing the maintenance and inspection, be sure to turn off the main power switch of the controller. • When disposing of batteries, be sure to follow the user's regulations.

! CAUTION	
 Disassembly prohibited	• The user should NEVER replace or change parts other than those stipulated in the instruction manual. Otherwise, the performance will deteriorate, resulting in troubles.
 Mandatory	• To replace parts, use the spare parts designated by SHIBAURA MACHINE. • Carry out the maintenance and inspection on a regular basis. Otherwise, the equipment may go wrong or accidents will be caused.

! CAUTION	
	To perform the maintenance and inspection of the robot with safety, read through and through the Maintenance Manual provided separately before actually starting the work.

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1. Operation Flow

The operation flow to enable the field bus interface is given below.
In the field bus specification, in case of specification, the customer of the controller purchase advance towards the step of the setting of (5) master bureau since then in case of the confirmation in the connection with the (3) master bureau.

(1) Mounting the field bus ⋯ See Section 2 of this manual.

Mount the field bus you procured on the controller.
Uncover the controller and perform the manual work.
Before starting this work, be sure to turn the power off.
When you already specified the field bus at order entry of the robot, the bus is factory-set. Make sure of it.
To proceed with the next process, the controller should remain uncovered.

⇓

(2) Confirming the hardware setting ⋯ See Sections 2 and 3 of this manual.

Set the switches equipped on the printed board of the field bus you procured.
Before starting this work, be sure to turn the power off.

⇓

(3) Connecting with the master station ⋯ See Section 2 of this manual.

Connect the field bus terminal on the controller side with the field bus cable connecting to the master station.

⇓

(4) Setting the controller parameters ⋯ See Section 3 of this manual, the operation manual and user parameter manual.

Turn on the controller power, edit the [U18 FIELD BUS] parameter contained in the user parameter (USER.PAR) file, edit and set the address assigned to the controller, baudrate, etc., using the controller editor.
The user parameter becomes operative after the power is turned off once, then on again.
As the setting of the master station is yet to finish at this point of time, turn off the controller power after this setting has finished.

⇓

(5) Setting the master station ⋯ See Section 3 of this manual and the master station instruction manual.

Perform setting of the master station. The setting procedures differ with the master station to be used. Refer to the master station instruction manual you procured.
The configuration file of the field bus printed board on the robot controller side

should be downloaded from the HMS's homepage as necessary.

⇓

(6) Confirming the operation ··· See Section 5 of this manual and the master station instruction manual.

To make sure that the I/Os assigned to the robot controller can be operated properly, set ON the switch assigned from the master station and monitor the I/O status from the robot controller. Also, set ON the switch on the robot controller side and monitor the I/O state from the master station to make sure that the information can be transmitted legally.
If an error has generated, investigate the error information from the master station. It is also possible to confirm the error state from the LED on the field bus printed board.

⇓

(7) Creating the robot program and verifying the control ··· See Section 6 of this manual and the master language instruction manual.

Create the robot program, referring to the example of robot program creation and make sure that the I/Os on the field bus can be controlled, using the SCOL language.

⇓

(8) Finish of the work

Now the work has finished. Turn off the controller power and cover the controller.

2. Hardware Structure

2.1 AnyBus Printed Board Assembling Procedures

Assemble the AnyBus printed board in the manner described below. (See Fig. 1.)

a) Provide the X8HL printed board, AnyBus cable, HL–CN plate, PWB set plate, spacer and various screws.

b) Attach the CN–HL plate to the X8HL printed board from the side A, using two (2) screws (M3 × 6).
(The HL–CN plate comes in the two (2) types; for DeviceNet and CC–Link, and for Profibus. (See Fig. 2.)

c) Attach the PWB set plate to the X8HL printed board from the side B, using one (1) screw (M3 × 6).

d) Set the three (3) M2.6×7 spacers on the side B of the X8HL printed board, and connect the AnyBus printed board to X8HL–CN2, using six (6) screws (M2.6 × 6).

e) Insert the AnyBus cable from the front side (i.e., side SYLK-displayed) of the notched hole on the CN–HL plate.
- For the DeviceNet and CC–Link, insert the cable while the connector [2] is disconnected, then connect the connector [2]. The connector [2] is attached when you bought the AnyBus printed board.
- For the Profibus, the cable can be inserted with the connector [2] connected.

f) Secure the connector [1] to the HL–CN plate.
- DeviceNet and CC–Link ⋯ M3 × 6 cross-recessed pan head screw × two (2) pcs.
- Profibus ⋯ OMRON-make special spacer XM2Z–0012 × two (2) pcs.

g) Attach and connect the connector [2] to the AnyBus printed board.

h) Affix the AnyBus function name seal to the HL–CN plate. (For the affixing position, see Fig. 2.)
Now, the AnyBus printed board has been assembled.

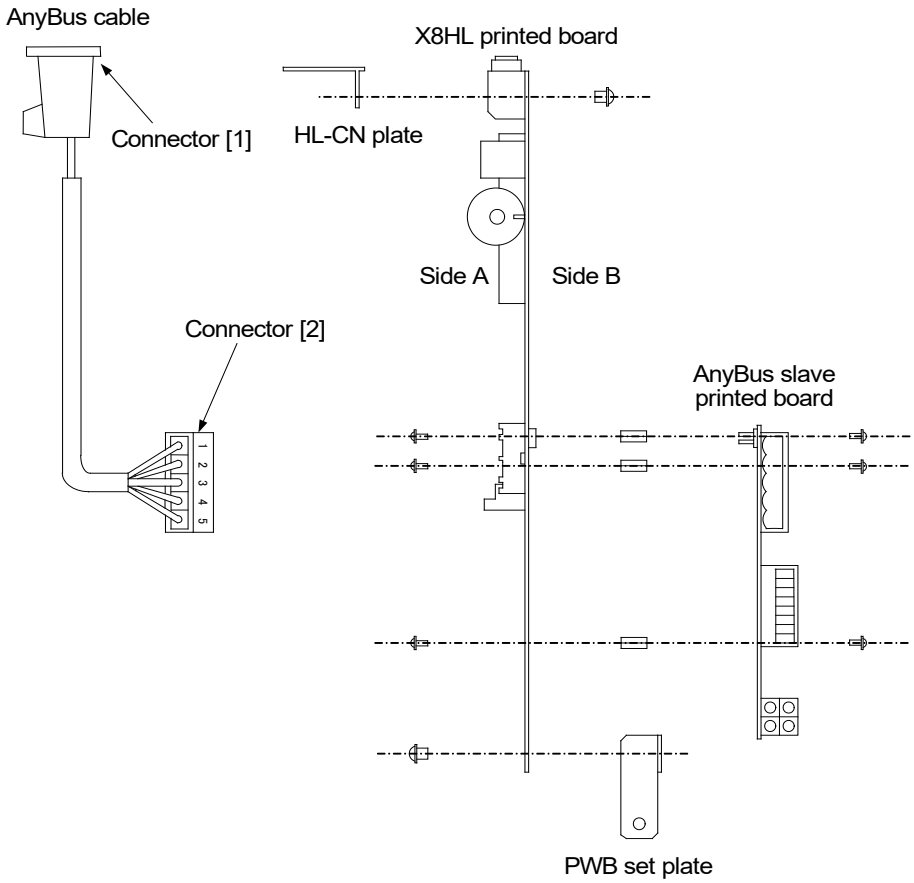
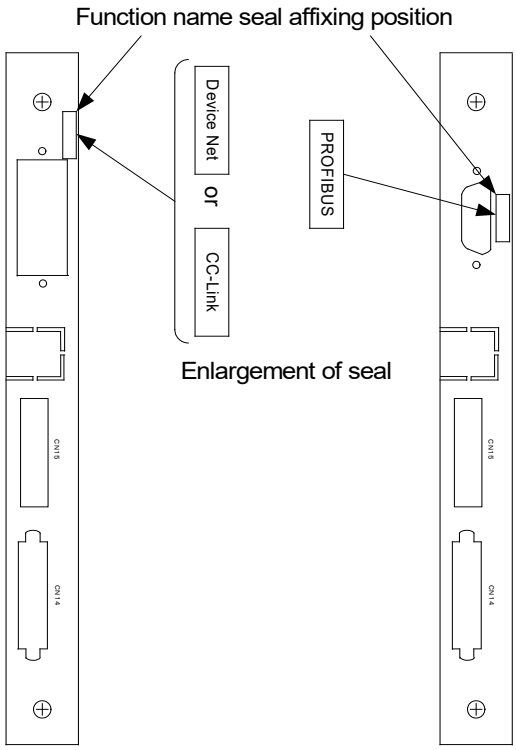


Fig. 1 AnyBus printed board assembly drawing



(a) For DeviceNet and CC–Link (b) For Profibus

Fig. 2 HL–CN plate

2.2 X8HL Printed Board Assembling Procedures

Assemble the X8HL printed board to the robot controller according to the following procedures.

a) Turn off the power of the robot controller.

b) Turn off the main power of 200 V supplied to the robot controller.
Disconnect the power connector (ACIN–CN1) to assure safety.

c) Remove the top cover of the robot controller.
This top cover is secured with a total of nine (9) screws (M3 × 6) (four (4) screws each on the both lateral sides and one (1) screw on the top side of the robot controller. Remove the screws, using a screwdriver, then draw the cover toward the front side and remove.

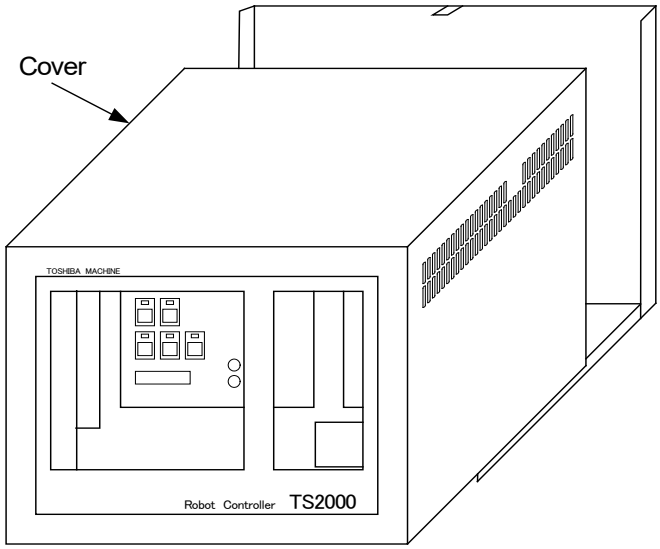


Fig. 3 Removing the cover

d) Remove the dummy connector plate attached to the rear side of the robot controller. (This plate is secured with two (2) M3×6 screws.)

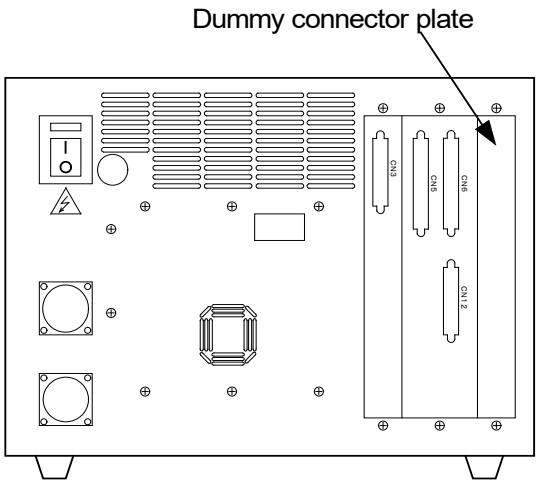


Fig. 4 Rear side of robot controller

e) Insert the X8HL printed board equipped with the AnyBus printed board into the mother board.
Remove the connector cover of the connector CN6 on the mother board (X8HB), then completely insert the X8HL printed board to CN6 of the X8HB printed board.
After the insertion, secure the rear side (two (2) places) and internal steel plate (one (1) place) of the robot controller with the screws (M3 × 6) to secure the board.

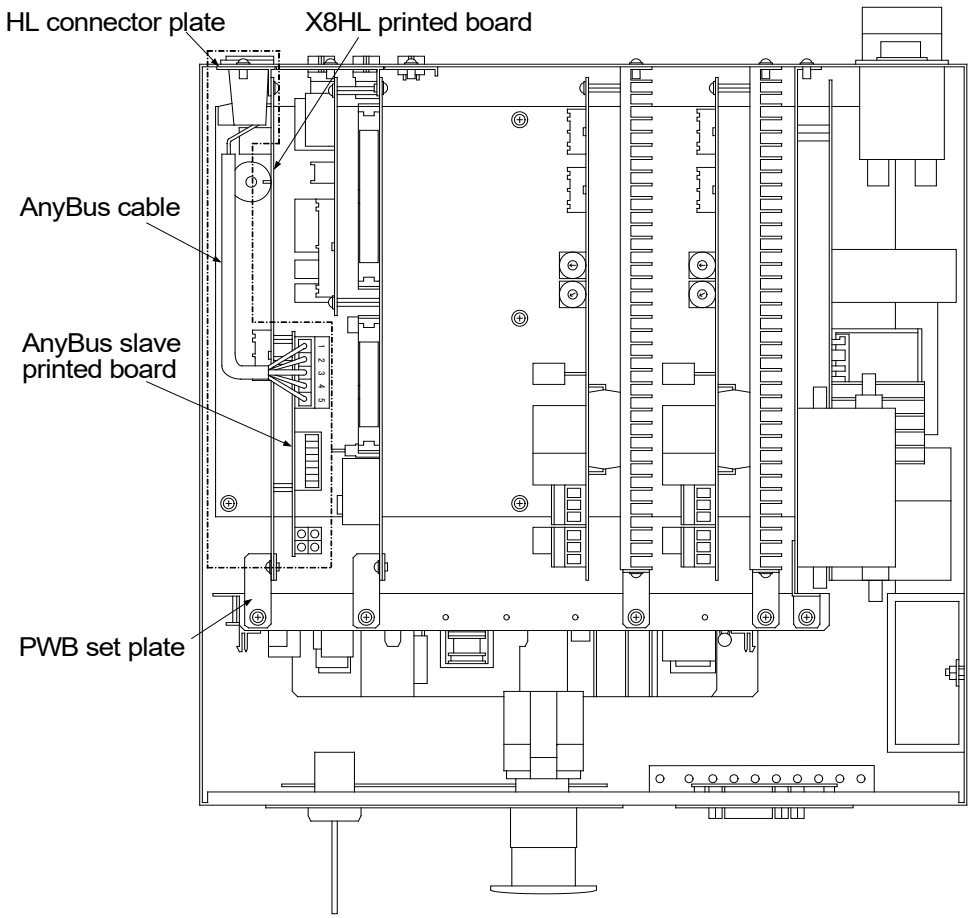
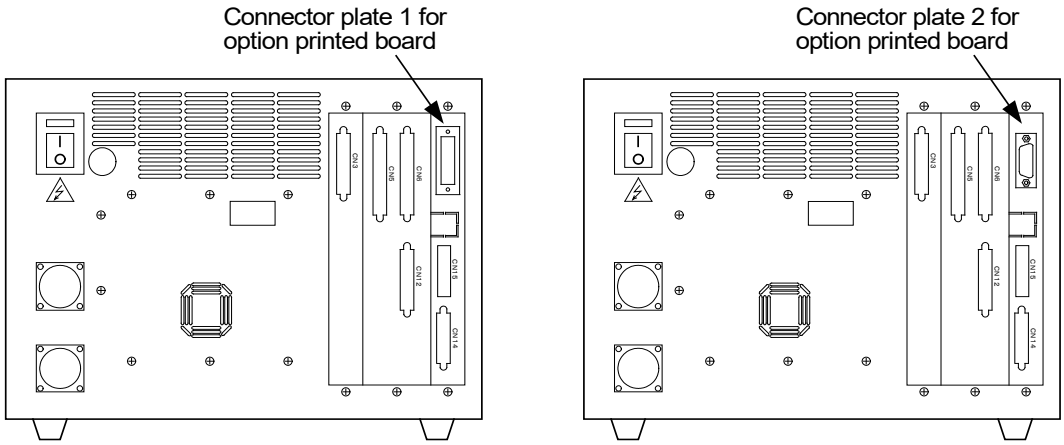


Fig. 5 Layout in robot controller panel

f) Attach the cover to the robot controller as originally set, then secure it with the screws.



(a) Rear side for DeviceNet and CC–Link (b) Rear side for Profibus

Fig. 6 Rear view of robot controller after assembly of printed boards

2.3 Connection

2.3.1 Pin Array of Connector for Connecting External Equipment

The pin array of the AnyBus cable in the controller panel is as shown below.

(a) DeviceNet

Pin No.	Signal name	Application
1	24G (V–)	Negative supply voltage
2	CAN_L	CAN_L bus line
3	SHIELD	Cable shield
4	CAN_H	CAN_H bus line
5	P24V (V+)	Positive supply voltage

Interface connector : IC2.5/5–STGF–5.08 made by Phoenix
Frame : IC–DFR5 made by Phoenix
Cable used in the panel : 1485C–P1C50 made by Rockwell, or equivalent

(b) Profibus

Pin No.	Signal name	Application
Housing	SHIELD	Connected to PE.
1	—	Not connected.
2	—	Not connected.
3	RXD (B-line)	Positive RxD/TxD according to RS485 specifications.
4	RST	Request to send
5	GND (V–)	Isolated GND from RS485 side.
6	P5V (V+)	Isolated P5V from RS485 side.
7	—	Not connected.
8	TXD (A-line)	Negative RxD/TxD according to RS485 specifications.
9	—	Not connected.

Interface connector : XM2A–0901 made by OMRON
Cable used in the panel : U–TKVVBS AWG20 3P made by Tachii W, or equivalent

(c) CC–LINK

Pin No.	Signal name	Application
1	DA	Communication line (DA)
2	DB	Communication line (DB)
3	DG	Digital GND (DG)
4	SHIELD	Cable shield
5	—	Frame ground (FG/PE)

Interface connector : IC2.5/5–STGF–5.08 made by Phoenix
Frame : IC–DFR5 made by Phoenix
Cable used in the panel : FANC–110–SBH 3X0.5 made by Kuramo, or equivalent

2.3.2 Connection

The reference connection drawing is shown in Fig. 7 below.
For the cable connecting the external equipment, use the recommended cable of the master sequencer (PLC) you use.

Use the following cable for the connector on the cable side.

(a) DeviceNet : MSTB2.5/5–STF-5.08
(b) Profibus : XM2D–0901 made by OMRON
(c) CC–Link : MSTB2.5/5–STF-5.08

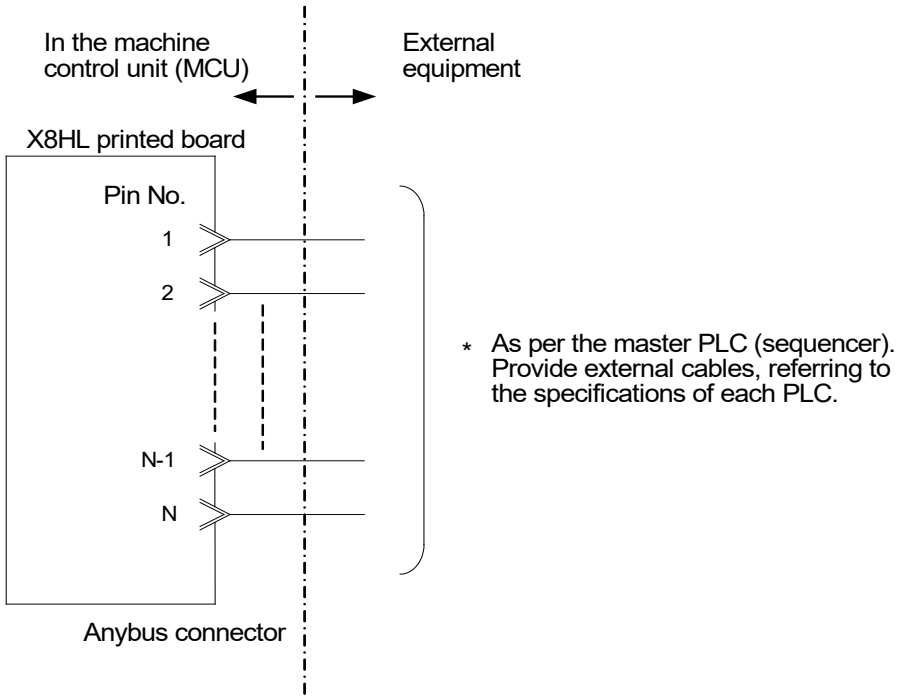


Fig. 7 Connection drawing for reference

3. Field Bus Slave Module

3.1 Profibus Slave Module Specifications

3.1.1 Profibus Specifications

Field bus specifications	Profibus
Master operation	Not supported.
Master device	16-bit little endian device (e.g., Siemens S7 315DP2) 16-bit big endian device
Node address	Both of the two (2) rotary switches equipped on the printed board should be set to zero (0). The node address is specified by user parameter (0 ~ 127).
Baudrate	Automatic detection (9.6 kbps, 19.2 kbps, 45.45 kbps, 93.75 kbps, 187.5 kbps, 500 kbps, 1.5 Mbps, 3 Mbps, 6 Mbps, 12 Mbps)
GSD file	Each device on the Profibus network relates to the GSD file containing all required information on the devices. The GSD file is used to establish the network. *1: The GSD file should be downloaded from the HMS's internet web site. Type: GSD Configuration File Name: AnyBus–S Profibus–DP
Number of inputs	64 numbers (DIN301 ~ DIN364)
Number of outputs	64 numbers (DOUT301 ~ DOUT364)

*1: HMS's internet web site

Homepage written in Japanese

http://www.hms-networks.co.jp/

Homepage written in English

http://www.hms-networks.com/index.html

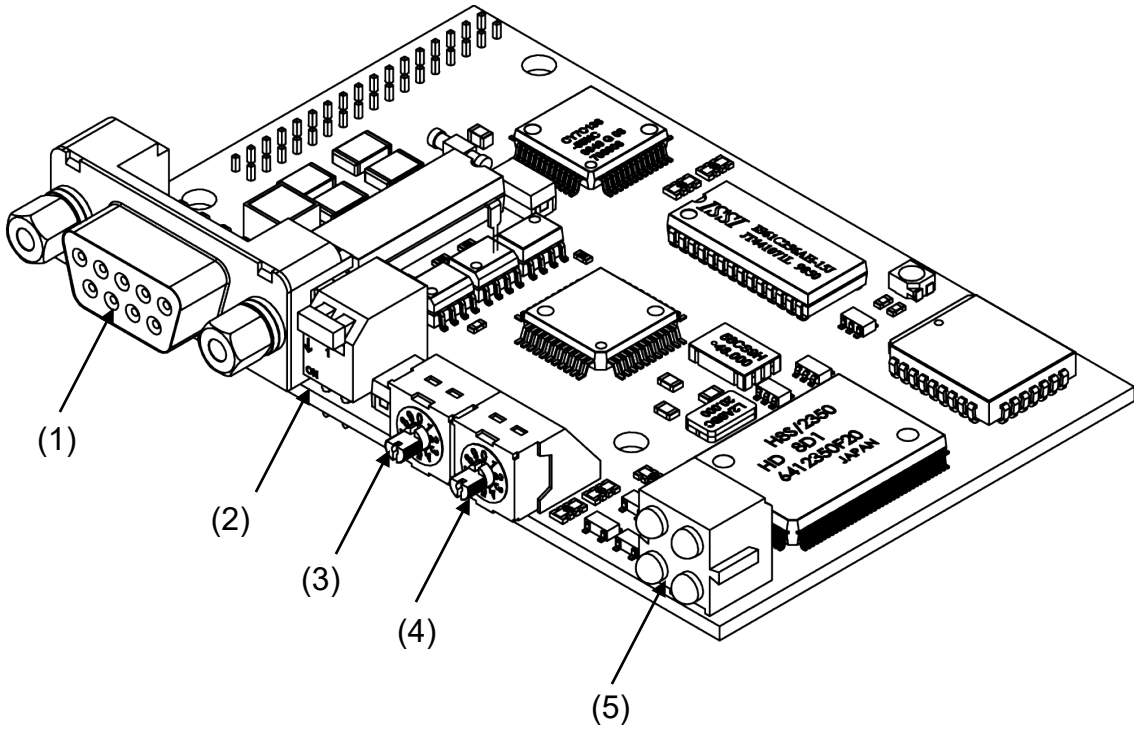
Japanese download site

http://www.anybus.com/jpweb/downloads/absdownloads.shtml

English download site

http://www.hms-networks.com/downloads/absdownloads.shtml

3.1.2 Profibus Connector, Indicator and Switch

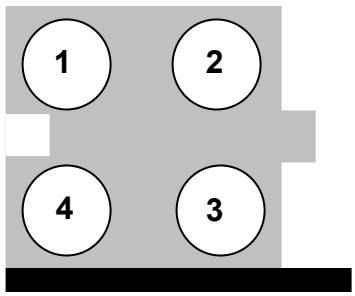


(1) Connector
Used to connect the Profibus.

(2) Terminator switch
Used to set ON or OFF of the bus terminator. When the module is used as the node 1 or final node on the network, the switch should be set ON. Otherwise, set it OFF. When the external terminator connector is used, set the switch to OFF.

(3) Rotary switch for higher value of node address
(4) Rotary switch for lower value of node address
These two (2) rotary switches should be set to zero (0).

(5) Status display
Used to confirm the operating condition by the four (4) LEDs.
The LED1 is illuminated in green as long as the field bus printed board is operating.
The LED2 is illuminated in green while this module is set online on the field bus.
In the clear mode, the LED flickers in green.



The LED3 is illuminated in red while this module is set offline on the field bus. The LED 4 flickers in red when an abnormality has occurred on the field bus side. When it is flickering at 1 Hz, the configuration data is illegal. Likewise, when it is flickering at 2 Hz or 4 Hz, the parameter value is illegal or an error has occurred during initialization of Profibus communication ASIC, respectively.

3.1.3 Setting of Profibus User Parameter

The Profibus user parameter should be specified, using [U18] of the USER. PAR file. The contents of [U18] are shown below.

= (Type of field bus) (Node) (Baudrate) (Type of master)

```
[U18] FIELD BUS
= –1   –1   0   0
```

(Type of field bus) : Specify the type of the field bus.
Specify "1" for the Profibus.
–1 : Without any field bus option (initial value)
1 : Profibus

(Node) : Specify the node address of the field bus.
Take careful precautions not to share the same address with any other device in the same network.
The range of the set value differs with the type of the field bus.
–1 : Without any field bus option (initial value)
0 ~ 127 : Effective range of Profibus node address

(Baudrate) : Specify the baudrate.
Profibus baudrate is set from the master side. As the robot controller automatically detects it, the baudrate need not be specified on the robot controller side.

(Type of master) : Specify the type of the field bus master.
As the byte order (endian) differs with the master, specify the type according to the master.

0 : 16-bit big endian
1 : 16-bit little endian (S7 315DP2 made by Siemens)
2 : 32-bit big endian
3 : 32-bit little endian

Example setting 1:
When the Profibus is selected, the node address is 2 and Siemens's S7 316DP2 is used as the master module:

[U18] FIELD BUS
{Type / Node Addr / Speed / Byte Order}
= 1 2 0 1

Example setting 2:
When the field bus slave module is neglected:

[U18] FIELD BUS
{Type / Node Addr / Speed / Byte Order}
= –1 –1 0 0

3.2 DeviceNet Slave Module Specifications

3.2.1 DeviceNet Specifications

Field bus specifications		DeviceNet
Master/scanner operation		Not supported.
Master device		16-bit little endian device 16-bit big endian device (e.g., OMRON DRM21)
Node address		An exclusive node address is assigned to the DeviceNet on the network. The node address consists of 0 ~ 63, which is used to identify each node. The node address is specified by user parameter (0 ~ 63).
Baudrate		The baudrate is specified by user parameter. 0: 125 kbps 1: 250 kbps 2: 500 kbps
EDS file		Each device on the DeviceNet network relates to the EDS file containing all required information on the devices. The EDS file is used to establish the network. *1: The EDS file should be downloaded from the HMS's internet web site. Type: EDS Configuration File Name: AnyBus–S DeviceNet Parallel
Explicit message		Not supported.
I/O message		Not supported.
Predefined Master/SlaveConnectionSet		
	Polling	Supported.
	Bit strobe	Supported.
	Cyclic	Supported.
	Change of state	Supported.
	Number of inputs	64 numbers (DIN301 ~ DIN364)
	Number of outputs	64 numbers (DOUT301 ~ DOUT364)

*1: HMS's internet web site

Homepage written in Japanese

http://www.hms-networks.co.jp/

Homepage written in English

http://www.hms-networks.com/index.html

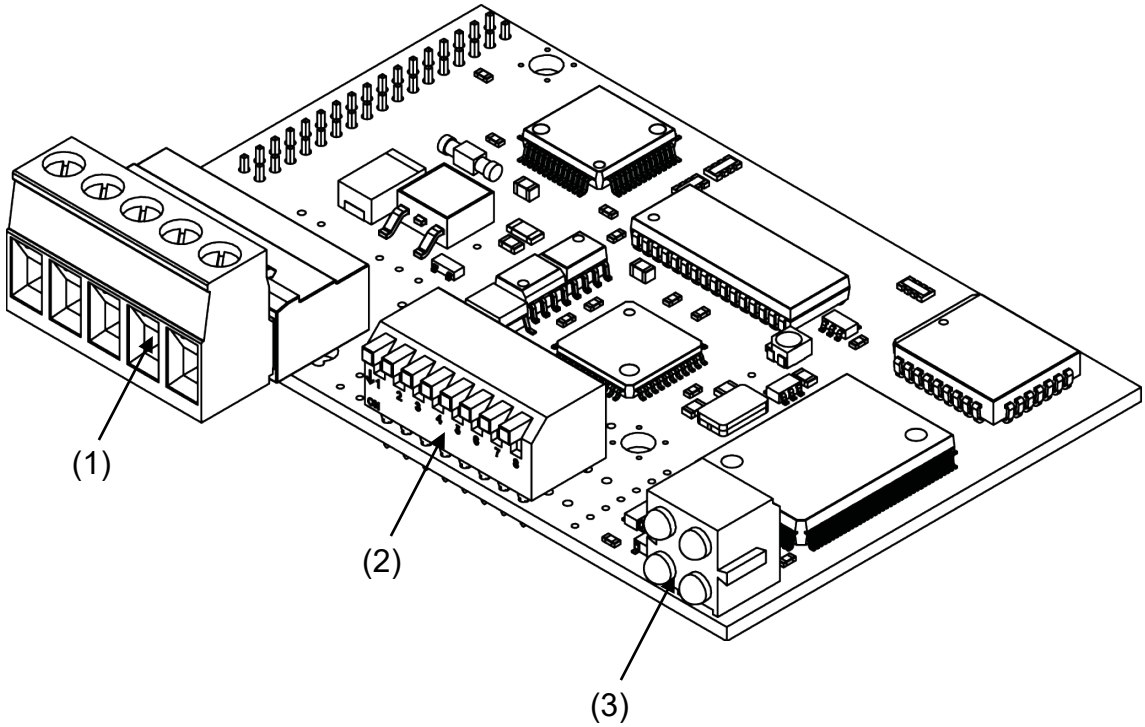
Japanese download site

http://www.anybus.com/jpweb/downloads/absdownloads.shtml

English download site

http://www.hms-networks.com/downloads/absdownloads.shtml

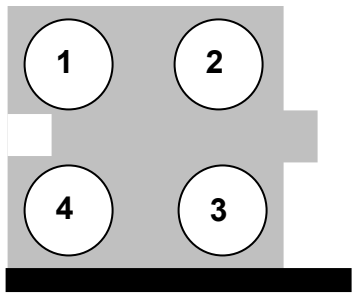
3.2.2 DeviceNet Connector, Indicator and Switch



(1) Connector
Used to connect the DeviceNet.

(2) Baudrate and node address DIP switch
As the software setting by user parameter takes priority, the DIP switch need not be set.

(3) Status display
Used to confirm the operating condition by the four (4) LEDs.
The LED1 is reserved for the system and is not used.
The LED2 is illuminated in green while this module is set online on the field bus.
While it is flickering in green, the module is set online, but is not connected with the master module. If it is illuminated in red, the link error has occurred. If it is flickering in red, the communication timeup error has occurred.



The LED3 is illuminated in green while this module is operating normally. If it is flickering in green, the data size is larger than the value set at the time of initialization. If it is illuminated in red, a serious failure that cannot be restored has occurred. If it is flickering in red, a minor failure that can be restored has generated.
The LED 4 is reserved for the system and is not used.

3.2.3 Setting of DeviceNet User Parameter

The DeviceNet user parameter should be specified, using [U18] of the USER. PAR file. The contents of [U18] are shown below.

= (Type of field bus) (Node) (Baudrate) (Type of master)

```
[U18] FIELD BUS
= –1   –1   0   0
```

(Type of field bus) : Specify the type of the field bus.
Specify "37" for the DeviceNet.
–1 : Without any field bus option (initial value)
37 : DeviceNet

(Node) : Specify the node address of the field bus.
Take careful precautions not to share the same address with any other device in the same network.
The range of the set value differs with the type of the field bus.
–1 : Without any field bus option (initial value)
0 ~ 63 : DeviceNet

(Baudrate) : Specify the baudrate according to the master baudrate.
0 : 125 kbps
1 : 250 kbps
2 : 500 kbps

(Type of master) : Specify the type of the field bus master.
As the byte order (endian) differs with the master, specify the type according to the master.

0 : 16-bit big endian (DRM21 made by OMRON)
1 : 16-bit little endian
2 : 32-bit big endian
3 : 32-bit little endian

Example setting 1:
When the DeviceNet is selected, the node address is 2, the baudrate is 125 kbps and OMRON DRM21 is used as the master module:

[U18] FIELD BUS
{Type / Node Addr / Speed / Byte Order}
= 37 2 0 0

Example setting 2:
When the field bus slave module is neglected:

[U18] FIELD BUS
{Type / Node Addr / Speed / Byte Order}
= –1 –1 –1 –1

3.3 CC–Link Slave Module Specifications

3.3.1 CC–Link Specifications

Field bus specifications	CC–Link Version 1.0
Master operation	Not supported.
Master device	16-bit little endian device 16-bit big endian device (e.g., MELSEC A1SJH)
Node address	Both of the two (2) rotary switches equipped on the printed board should be set to "9" The node address is specified by user parameter (1 ~ 64).
Baudrate	The rotary switch equipped on the printed board should be set to "9". The baudrate is specified by user parameter. 0: 156 kbps 1: 625 bps 2: 2.5 Mbps 3: 5 Mbps 4: 10 Mbps
CSP file	Each device on the CC–Link network relates to the CSP file. The CSP file is used to establish the network. *1: The CSP file should be downloaded from the HMS's internet web site. Type: CSP Configuration File (OS3) Name: AnyBus–S CC–Link Station 3
No. of occupied stations	Three (3) stations
Number of inputs	64 numbers (DIN301 ~ DIN364)
Number of outputs	64 numbers (DOUT301 ~ DOUT364)

*1: HMS's internet web site

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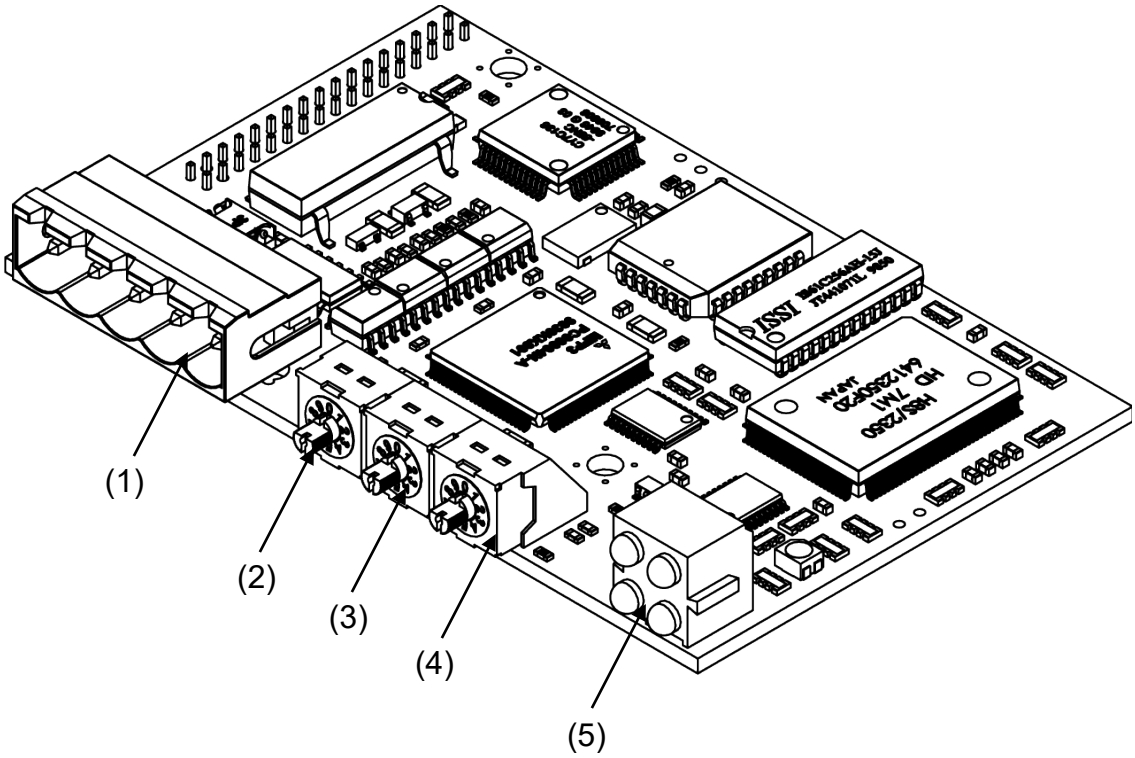
Japanese download site

http://www.anybus.com/jpweb/downloads/absdownloads.shtml

English download site

http://www.hms-networks.com/downloads/absdownloads.shtml

3.3.2 CC–Link Connector, Indicator and Switch



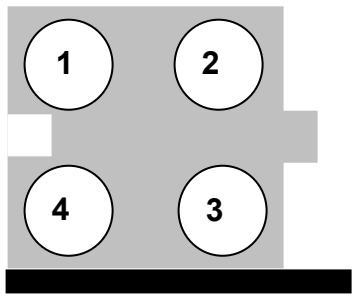
(1) Connector
Used to connect the CC–Link.

(2) Baudrate setting rotary switch
This rotary switch should be set to "9".

(3) Rotary switch for higher value of node address

(4) Rotary switch for lower value of node address
These two (2) rotary switches should be set to "9".

(5) Status display
Used to confirm the operating condition by the four (4) LEDs.
The LED1 is illuminated in green as long as the field bus printed board is operating normally.
The LED2 is illuminated in red if the CRC error has occurred, or if an illegal station number or illegal baudrate is selected.



The LED3 is illuminated in green when the data has reached.
The LED4 is illuminated in green when the data has been sent.

3.3.3 Setting of CC–Link User Parameter

The CC–Link user parameter should be specified, using [U18] of the USER. PAR file. The contents of [U18] are shown below.

= (Type of field bus) (Node) (Baudrate) (Type of master)

```
[U18] FIELD BUS
= –1   –1   0   0
```

(Type of field bus) : Specify the type of the field bus.
Specify "144" for the CC–Link.
–1 : Without any field bus option (initial value)
144 : CC–Link

(Node) : Specify the node address of the field bus.
Take careful precautions not to share the same address with any other device in the same network.
The range of the set value differs with the type of the field bus.
–1 : Without any field bus option (initial value)
1 ~ 64 : CC–Link

(Baudrate) : Specify the CC–Link baudrate according to the master baudrate. Setting is unnecessary for the Profibus.
0 : 156 kbps
1 : 625 kbps
2 : 2.5 Mbps
3 : 5 Mbps
4 : 10 Mbps

(Type of master) : Specify the type of the field bus master.
As the byte order (endian) differs with the master, specify the type according to the master.

0 : 16-bit big endian (A1SJH made by MELSEC)
1 : 16-bit little endian
2 : 32-bit big endian
3 : 32-bit little endian

Example setting 1:
When the CC–Link is selected, the node address is 2, the baudrate is 156 kbps and MELSEC's A1SJH is used as the master module:

[U18] FIELD BUS
{Type / Node Addr / Speed / Byte Order}
= 144 2 0 0

Example setting 2:
When the field bus slave module is neglected:

[U18] FIELD BUS
{Type / Node Addr / Speed / Byte Order}
= –1 –1 –1 –1

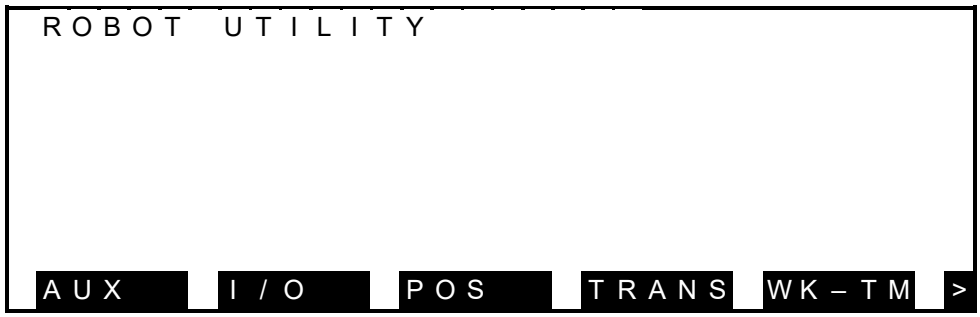
4. Error Detection

If an error has been detected during the initial parameter setting, the "008–353 Fieldbus Parameter Error" is output, which cannot be reset unless the power is turned off, then on again.
If an error has been detected on the field bus network, the field bus module enters an offline status with the "008–354 Fieldbus Offline" error being output. If the robot is moving then, emergency stop is effected on it.
This error continues until the module changes over to the online status.
The details causing the offline status should be investigated on the side of the field bus master equipment.

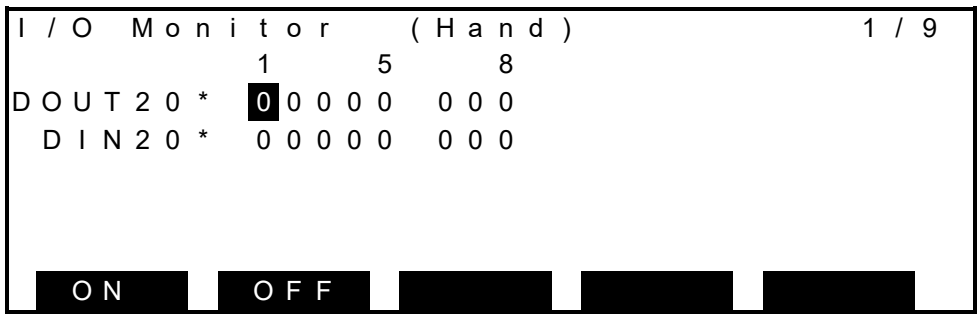
5. Verifying Motion

The field bus input signals are assigned to DIN301 through DIN364.
The field bus output signals are assigned to DOUT301 through DOUT364.
Respective motions of the above signal cables can be confirmed in the utility mode.
For detailed operation of the utility mode, see the operation manual provided separately.
To verify the motion of the signal cable, observe the following procedures.
Make sure that the setting of the user parameter and field bus master have finished, and that no error is generated in the field bus or controller.

Press the UTILITY key provided on the teach pendant, and the utility mode is selected and the system calls the following screen on the display.



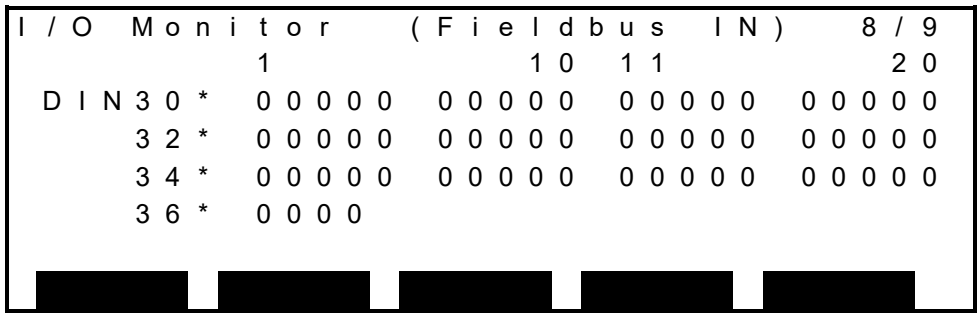
Press the F2 key (I/O) on the teach pendant, and the external input/output signal display appears. The following hand input/output signal status display is shown.



Press the NEXT key or the "ALT + ⇓" keys seven (7) times to call the Page 8 where the inputs from the field bus are displayed.
Now, you can make sure that the sixty-four (64) field bus input signals assigned from the master side can be turned on and off.

The input or output signal status is "1" when the contact is open. It is "0" when the contact is closed. A total of twenty (20) signals, separated by every five (5) signals, are displayed on the same line with the leading signal number shown on the left end.

Page 8 (Field bus input)



Press the NEXT key or the "ALT + ⇓" keys to call the Page 9 where the outputs from the field bus are displayed.
Now, you can make sure that the sixty-four (64) field bus output signals can be turned on and off.
The operative keys and contents of operation are listed below.

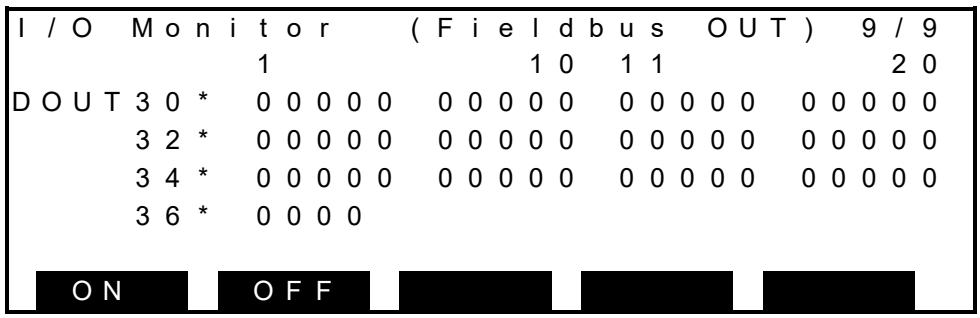
Key	Descriptions
←	Used to move the cursor to the left. When the cursor is located at the left end, it will not move any further.
→	Used to move the cursor to the right. When the cursor is located at the right end, it will not move any further.
⇑	Used to move up the cursor. When the cursor is located at the top, it will not move any further.
⇓	Used to move down the cursor. When the cursor is located at the bottom, it will not move any further.
ALT + ⇓	Used to call the next page. When the final page is displayed, the page will not change over.
ALT + ⇑	Used to call the previous page. When the top page is displayed, the page will not change over.
ESC	Used to return the system to the utility screen.

The operative soft keys and contents of operation are listed below.

Key	Descriptions
[ON]	Turns on the bit indicated by the cursor.
[OFF]	Turns off the bit indicated by the cursor.

The input or output signal status is "1" when the contact is open. It is "0" when the contact is closed. A total of twenty (20) signals, separated by every five (5) signals, are displayed on the same line with the leading signal number shown on the left end. You can make sure that the signals assigned from the master side can be turned on and off.

Page 9 (Field bus output)



6. How to Use SCOL Language Instruction

The field bus signals can be operated by the SCOL language instructions.
For details on how to use each instruction of the SCOL language, see the robot language instruction manual.

DIN	:	Reads the specified input signal status.
DOUT	:	Outputs the specified output signal.
RESET DOUT	:	Turns off the user's output signal.
PULOUT	:	Outputs the specified output signal in 0.2 second-interval pulse. The output time by the PULOUT instruction to this output area may be longer than 0.2 second, the normal DOUT output time.
BCDIN	:	Reads in BCD code the input signals whose length is four (4) times the specified signal length, starting from the specified signal.
BCDOUT	:	Converts the value of the expression into the BCD code and outputs the value equivalent to the number of digits as specified by the signal length to the output signals whose length is four (4) times the specified signal length, starting from the specified signal.
HEXIN	:	Reads in HEX code the input signals whose length is equivalent to the specified signal length, starting from the specified signal.
HEXOUT	:	Converts the value of the expression into the HEX code and outputs the value equivalent to the number of digits as specified by the signal length to the output signals whose length is equivalent to the specified signal length, starting from the specified signal.

6.1 Sample Program

This is the program in which the robot waits until input signal 301 turns on, then moves to taught points A1, A2 and A3.

```
PROGRAM  DINSAMPLE
WAIT DIN(301)
MOVE A1
MOVE A2
MOVE A3
END
DATA
POINT A3        =  1500.000,   0.000,   0.000,   0.000,   0.000 /
RIGHTY
POINT A2        =  1500.000,   0.000,   0.000,   0.000,   0.000 /
RIGHTY
POINT A1        =  1500.000,   0.000,   0.000,   0.000,   0.000 /
RIGHTY
END
```

This program turns on and off output signals 301 through 316 in turn.

```
PROGRAM  DOUTSAMPLE
FOR K=301 TO 316
DOUT(K)
TIMER=0.5
WAIT TIMER==0
DOUT(-K)
NEXT K
END
```

Appendix A Extending the Numbers of Profibus IOs to 128/128

Only when the Profibus is selected, the numbers of I/Os can be increased to 128/128 by the following setting. To use the I/Os thus increased, you have to create a user sequence program by using the simplified PLC function.

1. Parameter setting
   Specify number "1" for the 5th function on the 1st line of user parameter [U29].

   [U29] Function select switch
   This parameter selects ON or OFF of the controller functions.
   The functions which can be effective are assigned to each element.

   1 : Function ON
   0 : Function OFF

   [U29] Function select switch
   = 0 0 0 0 0 0
   = 0 0 0 0 0 0
   = 0 0 0 0 0 0
   = 0 0 0 0 0 0

2. Assigning increased I/O contacts to simplified PLC
   A total of 64 increased DIs are assigned to X24 W ~ X27W, and inputs from the Profibus are set there.
   Likewise, a total of 64 increased DOs are assigned to Y34 W ~ Y37W, and outputs to the Profibus are set there.

3. Adding contact output for designation of I/O extension
   When the I/O extension is designated, R518 (FL_EX) is set ON.